Diego Garcia

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Education

California Institute of Technology **Jun. 2025**
B.S. Mechanical Engineering *Pasadena, CA*

Relevant Coursework

Mechanical Prototyping, Design and Fabrication, Mechanics, Systems Engineering, Dimensional & Data Analysis, Thermal Science, Robotic Systems, Mobile Robots, Feedback Control, Aerospace Control, Linear Systems

Experience

Vacco Industries **Jul. 2025 – Present**
Product Development Engineer *Los Angeles, CA*

- Own project engineering for two valve programs transitioning from development to qualification, driving test planning, requirements verification, and design iteration to advance them toward production readiness
- Independently manage nonconformance dispositions and quality notifications for a production program, applying GD&T (ANSI/ASME Y14.5) analysis to resolve 25 dispositions/month while maintaining full compliance with customer and NAVSEA specifications.
- Perform engineering review of field-returned and legacy hardware, validating technician rework/scrap determinations and scoping NRE for heritage part drawing updates, on approximately 10 units monthly to identify cost savings and support accurate customer quoting.
- Author test procedures and technical data packages for SUBSAFE Level I submarine components, ensuring compliance with NAVSEA quality requirements and supporting government-witnessed First Article Inspections.
- Lead technical design reviews with defense customers at key program milestones, translating engineering specifications and compliance data into client-facing frameworks to secure customer approval to advance to the next design phase.

Gharib Research Group **Jan. 2024 – Jun. 2025**
Undergraduate Researcher *Pasadena, CA*

- Developed a bio-inspired morphing drone, validating thin-plate deformation models on a custom test rig by implementing and tuning a PID control algorithm to optimize flight telemetry.

The Aerospace Corporation **Jun. 2022 – Nov. 2022**
Research Fellow *Los Angeles, CA*

- Architected high-throughput, GPU-accelerated data pipelines in Java/CUDA to process high-volume, real-time telemetry feeds; optimized memory management and parallel processing constraints to slash latency by 15%.

Projects and Leadership

Autonomous Mobile Robot (Caltech ME/CS/EE 169) **May. 2025**

- Built and programmed an autonomous robot using ROS, Lidar, and sensor fusion for localization and mapping, designing predictive obstacle avoidance algorithms that placed 4th in a class-wide challenge.

Caltech Rover Autonomy, Technology and Exploration Research **Sep. 2022 – Jun. 2025**
Mechanical Co-Lead

- Co-led a multi-disciplinary team through the design and FEA validation of a 6-DOF robotic arm, successfully presenting technical validation data to stakeholders to accelerate integration timelines while writing winning grant proposals.

Autonomous Shuttle Bot (Caltech ME/CS/EE 75abc) **Mar. 2024**
Team Lead

- Spearheaded technical discovery and system architecture for a NASA Finalist vehicle, deploying a custom YOLO object detection model trained on hand-labeled datasets to enable real-time crater detection and autonomous terrain navigation at the edge (Jetson Orin Nano).

Technical Skills

Software & Frameworks: Git/Version Control, Python, Matlab, ROS, Java, C++, Linux, Bash, NumPy/Pandas, Edge Computing, PyTorch
Design and Systems Integration: CAD (SolidWorks, Catia, Fusion 360, AutoCAD, Inventor), GD&T, DFMA, FEA (ANSYS), Control Design, Hardware-in-the-loop-testing (HIL) Testing, PLM (Autodesk, SAP)
Hardware Prototyping: Additive Manufacturing, Mechatronics, Rapid Prototyping, Molding, Machining, Sensors and Actuators, Microcontrollers (Arduino, Teensy), Sensor Integration (encoders, IMU, LiDAR), Jetson/Raspberry Pi

Leadership / Extracurricular

Caltech NCAA Soccer **Sep. 2021 – Jun. 2025**
Team Member *Pasadena, CA*